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SUPERSEDING
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7 October 1981

MILITARY SPECIFICATION

SHEARING MACHINES, SHEET METAL, POWER DRIVEN

This specification is approved for use by all Departments and Agencies of the Departnmt of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of power driven mtal squaring -s for precision shearing of light gauge sheet metal.

1.2 Classification. The shearing machines shall be of the following types. The size shall be determined by material thickness and cutting length capacity. The type and size to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Gapless Fram

Type II - Gap Frame

2. APPLICABLE DOCUMENTS

2.1 Governmnt documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplemnt thereto, cited in the solicitation (see 6.2).

Beneficial comnents (recommendations, additions, deletlons) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipmnt Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Menphis, Tennessee 38114-5051, by using the self-addressed Standardization Docuwnt Improvement Proposal (DD Form 1426) appearing at the end of this documnt or by letter.

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SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Require~nts for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Un"less otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Docu&nts Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Governmmt documents, drawings, and publications. The following other Governmnt documnts, drawings, and publications form a part of this documnt to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIOtJAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documnts, U. S. Governwnt Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Governmnt documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Governmnt publications. The following documents form a part of this documnt to the extent specified herein. Unless otherwise specified, the issues of the docu~nts which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documnts cited in the solicitation (see 6.2).

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AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Itanufacturers' Association, Standards Departmnt, Suite 201, 1500 King Street, Alexandria, VA 22314 -2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B1l.4 - Shears - Safety Requirements for the Construction, Care, and Use. (DOD adopted)

(Application for copies should be addressed to the Amrican National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (1S0)

1S0 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Battery march Park, Quincy, MA 02269 -9101.)

NATIONAL FLUID POWER ASSOCIATION, INC. (NFPA)

NFPA T2.24.1 - Hydraulic Fluid Power - Systems Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fluid Power Association Inc., 3333 N. Mayfair Road, Milwaukee, WI 53222-3291.)

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(Non-Governmnt standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the t~"t of this documnt takes precedence. Nothing in this docunmt, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one conplete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The shearing machine shall be new and one of the manufacturer's current nmdels and capable of shearing material in accordance with " the requirements herein. The machine shall be a self-contained power driven unit and designed to accomplish shearing by the downward movemnt of a crosshead rmunted upper shear blade (knife) past a stationary lower shear bl ade. The machine's design and construction shall include all components, parts and features necessary to met the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustmnt, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1) or mtric (see 6.5.2), units as specified by the procuring activity (see 6,2). Regardless of the masurerm?nt system used, all measuring and indicating devices on the machine shall be graduated in t?e sarm systsm.

3.2.1.1 Dials. All dials or digital readouts used to indicate workp*ece* position shal~ graduated in incre~nts not larger than 0.001 inch. All controls shall be permanently and legibly engraved or etched on a nong are background that can be read from the operator's normal position.

3.2.2 Reclaimd materials. The machine may contain reclaimd materials to the maxirwm extent possible provided such materials will not jeopardize intended use, performance, or design life of the machine. Reclaimd materials hall have been collected or recovered from solid waste and reprocessed to becom a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to man that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for en*ergy* efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All electrical, mechanical, and hydraulic operating controls shall be located convenient to the operator's work stations.

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3.2.5 Safety and health requirements. All parts, components, mechanisms, and assenblles furnished on the machine, whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910 and ANSI B1l.4. If a conflict arises between OSHA 29 CFR 1910 and ANSI B1l.4, OSHA 29 CFR 1910 shall apply. Additional safety and health requirenmts shall be as specified (see 6.2).

3.2.5.1 Cows guards and other safety devices. *Covers,* guards, or other safetv devices normally furnished as standard on the manufacturer's cormrcial machifie supplied to th; commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration the procuring activity shall provide part design data to the supplier of the machine (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the operation of the machine. Al 1 guards shall allow access to the guarded parts.

3.2.5.2 Noise level. Unless otherwise specified (see 6.2), noise emitted by the machine when operating under a no load condition shall not exceed 84 db when m+asured on the A scale of a type 1 or type 2 sound level meter at slow response.

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury conpounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall *not* be used on or In the machine.

3.2.8 Environrmntal protection. During the manufacture, service, transportation, or storaqe of the machine, and its co~onents, the machine shall" meet alj applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. The operation of the machine at its final location shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission materials hazardous to the environment or the operator's health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. All oil holes, grease fittings, and filler caps shall be accessible. Unless otherwise specified (see 6.2), a centralized manual one-shot lubrication system with positive adjustable lubricant delivery rate, automatic high pressure relief mechanism, and cleanable or replaceable in-line filter shall be supplied and installed on the machine. Means for determining lubricant level in each reservoir shall be by sight gauge, warning light, or a similar device.

3.2.10 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustrmnt without nndification of parts or machine.

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3.3 Construction. The shearing machine shall be constructed of parts which are new, wltbout defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maxirrum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to rermval or adjustmnt shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's conmrcial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where speclfled In the original design. None of these operations shall be employed as a repair masure for any defective part.

3.3.5 Paintin Unless otherwise specified (see 6.2), the paint color and applica+Ion s all be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the shearing machine and its related attachnmts and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of the drive train shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2001.B88, and AGMA 2000-A88 for the inch-pound systemof measurement or 1S0 54 for the metric system of ~asuremnt. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipmnt furnished under this specification shall conply with MIL-STD-461. The equipmnt and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The shearing machine shall consist of not less than a frame, end housings, workpiece table, ram, holddown mechanism, machine drive, back gauge, front gauge, side squaring gauge, and shear blades.

3.4.1 Frame. The fram asserrt)ly shall support all components of the machi ne at a level of rigidity required to perform maximum shearing

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operations specified herein without permanent deformation or failure. The frame shall include two end housings, a workpiece table and an upper crosshead or cross m?nbers joined together to form a single unit. The frame metiers shall be constructed of either ductile iron castings or a welded steel fabrication. The mtiers shall be joined either with a bolted interlocking construction or as a stress relieved welded assefily. The frame shall support the top of the workpiece table at a workin! height of not less than 30 inches, and not me than 38 inches from the rmunt" ng surface.

3.4.2 End housin s. End housings sha' 1 be either ductile iron castings or steel pla ype I machine shal be provided with end housings of the gapless or straight design. End hous ngs for the type II machine shall conform to the requirenmts of table II to provide gaps or workpiece openings for obtainina off-cuts of not less than 10 inches wide from work~ieces of anY length. The-end housings shall be provided with guideways for supporting and guiding the ram and upper shear blade to accon-plish the shearing operations specified herein. The base of the end housings shall have means to permit leveling and fastening the machine to the floor or foundation.

3.4.3 Workpiece table. The workpiece table or bed shall be constructed of either ductile iron casting or steel plate, with the top surface finished to 125 micro inches or less, and flat within a tolerance of + 0.005 inch per linear foot. The off-cut edge of the workpiece table shall be provided with mans for securing and supporting the lower shear blade in a stationary shearing position. The workpiece table top shall be provided with holes, slots, or keyways, as required, for attaching and accorwrodating bevel gauges and side squaring gauges as specified herein. The machine shall be provided with an adjustmnt for obtaining and maintaining clearance alignmnt between the rmvable shear blade and stationary shear blade.

3.4.4 Ram. The ram shall be constructed of either a ductile iron casting or a fabri~ed steel plate (stress relieved), and designed to secure and guide the upper shear blade through the shearing cycle. The ram travel shall be acco~lished on non-metallic bearing surfaces with provision to compensate for wear'. Guide ways shall be prov" ded to guide the ram through the shearing action and to hold the upper shear blade in alignment. The guide ways shall be constructed of either a ductile " ron casting or tool steel and be directly supported by the frame.

3.4.5 Holddown mechanism. The machine shall be provided with a holddown mchanism ca~able of securln~ any s ze workpiece up to and includinq the rated capacity of the machine. Tie hblddown mchanisrn shall consist OF either a solid full width pressure bar or plunger type unit. The plunger type units shall consist of either spring loaded, hydraulic, or pneumatic units with a centerline spacing of not m-e than 12 inches across the full workpiece opening of the machine. The holddown mchanism shall be designed to provide an unobstructed view of the shearing action from the operator's stations. The plunger type units shall be equipped with holddown pads of resilient, oil-resistant non-scratch material to prevent damage to the workpiece.

3.4.6 Machine drive. The shearing action of the machine shall be accomplished by either an electric motor driven mchanical or hydraulic type drive. When only one type drive of the type I machine is acceptable, the

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machine is acceptable, the particular type drive required shall be as specified (see 6.2). The shearing action of the type II machine shall be accomplished by and electric motor driven mechanical drive. The machine shall be furnished with operating controls to provide both single non-repeat and continuous stroke operation. The operating controls shall include a foot control switch that shall be arranged to activate the machine drive when the starter switch is in the "power on" position.

3.4.6.1 Mechanical type drive system. Mechanically driven machines shall have an electric motor driven rmchanlcal drive system capable of initiating and controlling the reciprocating stroke of the ram. A dynamically balanced cast iron or steel flywheel shall supply supplemental power to the drive system. An eccentric shaft shall convert the rotary mtion of the motor and flywheel into the reciprocating nntion required by the ram to coq)lete the shearing stroke. The eccentric shaft shall rotate within bushings or anti-friction bearings and shall be directly supported by the end housings of the frame. The energy developed by the drive system shall be controlled by a clutch capable of transmitting the maximum developed torque to the eccentric shaft and ram. The machine shall be provided with a brake for stopping the motion of the ram at the top of the ram stroke when control is set for single stroke operation.

3.4.6.2 Hydraulic type drive system. Hydraulically driven machines shall be poviered by a hydraullc system that will accomplish the maxirrum shearing capacity as specified herein and to operate the workpiece hold down cylinders. The hydraulic type drive system shall conform to or exceed the requirements of NFPA T2.24.1 Hydraulic Fluid Power Standard and shall be conplete, including all pumps, valves, piping, cylinders, and pressure controls. The shear knife shall be reciprocated by a ram which is attached to one or nmre hydraulic piston and cylinder assetilies.

3.4.7 Gauges. The machine shall be provided with a back gauge, a front gauge, and two side squaring gauges. The gauges shall acconrmdate any workpiece width up to and including the rated capacity of the machine.

3.4.7.1 Back gauge. The back gauge shall provide an adjustable reference edge between the back edge of the workpiece and the lower shear blade. The back gauge shall be adjustable to any position within the back gauge capacity specified herein. The shearing machine shall be furnished with either a power or manually operated ball bearing parallel back gauge. The operating controls of both the power and manually operated back gauges shall be located on the front of the machine and shall be provided with a direct reading counter type position indicator that displays not nnre than 0.001 inch calibrations in the distance between the back gauge and the lower shear blade. A means of adjustmnt shall be provided for correcting any back gauge misalignment and adjustments shall be provided for taper cutting. When specified (see 6.2), the back gauge shall be furnished with either a magnetic or pneumatic workpiece support mchanism for supporting the off-cut portion of capacity width workpiece.

3.4.7.2 Front gauge. The front gauge shall consist of not less than two detachable support arms with adjustable work stops, to provide an adjustable reference edge between the front edge of the workpiece and the lower shear bl ade.

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3.4.7.3 Side squaring gauges. Both the right and left side of the machine shalT be provided with a detachable side squaring gauge for positioning the edge of the workpiece at right angles to the lower shear bl ade. Graduated scales calibrated in not rmwe than sixteenths of an inch and adjustable to compensate for blade wear shall be recessed in either the side squaring gauge or the surface of the bed to facilitate workpiece positioning. When specified (see 6.2), an extension side squaring gauge with length as specified shall be provided with the machine. The extension side squaring gauge shall be provided with a scale graduated in sixteenths of an inch and not less than two adjustable stops.

3.4.7.4 Light beam shearing gauge. When specified (see 6.2), the machine shall be provided with a light beam shearing gauge that utilizes light beam shadows at the shear line to facilitate shearing to a scribed line on the workpiece.

3.4.8 Shear blades. Shearing machines shall be provided with high carbon, high chrome tool steel shear blades. The shear blades shall be hardened to not less than 57 Rockwell C, and machined for endurance and accuracy. The shear blades shall be reversible, shall have four cutting edges, and shall be hardened and ground to co~ly with the manufacturer's standard practice. hfhen specified (see 6.2), an extra set of shear blades shall be furnished with the machine.

3.4.9 Electrical requirenmts. The shearing machine shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each shearing machine shall be designed to operate on 3-phase, 60-Hertz (Hz), dual 230/460 volts, Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The shearing machine shall be initially wired as specified (see 6.2).

3.4.9.1 Motors. Motors shall meet the requirements of 3.4.9. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) rotors shall be designed to operate on 60-Hz. The nmtor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.9.2 Solid state components. Solid state design shall be used throughout the *machine* for electronic co~onents.

3.4.10 Hour meter. When specified (see 6.2), the shearing machine shall be equipped with an hour inter. The hour mt~r shall be installed to display accumulated operating time of the main drive rotor. The inter shall be of the nonresetting type and shall have a range of O to 9,999 hours in increnmts of not greater than 1 hour. The inter shall be sealed to prevent the entrance of dust and misture and shall be nnunted to withstand shock and vibration generated by the shearing machine.

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3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the shearing machine shall meet the requirements of tables I and 11 for the machine specified. Machine size and capacity shall be not less - than the stated requirenwts of the size ordered. When a range is given, the required performance shall be from the stated minimm or less to the stated maxinum or greater.

3.5.1 Physical size limitations. When specified (see 6.2.), The physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Performance. The machine shall be capable of shearing mild steel of not less than 60,000 pounds per square inch (psi) tensile strength in workpiece sizes ranging up to the rated capacity for the thickness and length as specified in tables I and II. The rated capacity shall be sheared with the upper shear blade rake angle fixed, if applicable, or set at a rake angle not greater than specified in tables I and II. Shearing shall be accomplished with a shear blade clearance of not nnre than 15 percent of the workpiece thickness being sheared. The workpiece shall be free of visible hold-down cylinder damage.

3.6.1 Shearing accuracy. The machine, while shearing mild steel of not less than 60,000 PSI tensile strength, shall be capable of shearing maxinum thickness and up to the full width capacity of the machine to precision tolerances as follows:

a. Dimensions - The width of the sheared workpiece in relation to the - back gauge setting, shall be within ±0.001 inch per foot of the shear cutting length.

b. Parallelism - The sheared cut shall be par~llel to the sheared (trinrned) edge positioned against the back gauge within ~0.001 inch per foot of the shear cutting length.

3.6.2 Additional performance requirerm?nts. Performance features or shearing accuracies required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipmnt. All standard equipnnt normally provided with the manufacturer's conmrcfal shearing machine shall be furnished.

3.8 Optional and accessory equipmnt. Optional and accessory equipmnt shall be furnished as speclfled and fully described (see 6.2). The equipmnt provided shall be functional without requiring modification of the equiprmnt or machine.

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

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TABLE 1. Type 1, shearing machine, gdpless frame, sizes and capacities.

Machine Size	Capacities: Workpiece thickness (gauge)	Workpiece width [inches)	Rk A 1 (i~c~es~e; foot)	Shear Speed (strokes per minute)	Back Gauge capacity (inches)	t rent Gauge capacity (inches)
5	16	48	5/16	160	24	36
6	14	48	5116	80	24	36
7	14	52	7/16	25	24	36
8	14	72	5/16	80	24	36
9	14	120	3/8	60	24	36
10	12	24	318	85	24	36
11	12	52	3/8	25	24	x
12	12	72	318	60	36	42
13	10	48	5/16	60	36	42
14	10	72	3/8	60	36	42
15	10	96	318	60	36	42
16	10	120	3/8	50	36	42

TABLE II. Type 11, she~ring machine, gap frame, sizes ~nd capacities.

Machl ne Size	Capdcltles: Workpiece thickness (gauge)	Workpiece width (inches)	Rake Angle (inches per foot)	Shear Speed (strokes per minute)	Back gauge capacity (inches)	Gap, minimum (inches)
1	10	48	5/16	120	24	10

NOTE: Requirements sh~ll be not less than the figures shown except for the rake dngle ~ which sh~ll be not more th~n the figure shown.

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3.10 Marking on instruments, control panels, charts, and plates. All words on the machine lncludlng all instruments, control panels, control units, charts, plates, and related attachnmts shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate or chart. A co~rosion resistant lubrication plate shall be securel, Lttdched to each mach,ir ?. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Points of lubricant application
Type of lubricant
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

3.10.2 Namplate. A corrosion resistant nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the name-plate shall contain the information listed below:

Nomenclature
Manufacturer's nam
Manufacturer's mdel designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Anp rating of largest rmtor
Short circuit/over-current protection rating
Contract number or order nunber
National stock nunber or plant equipmnt code
Date of manufacture

3.10.3 Hydraulic fluid requirenr?nts plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hy-draulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Type of fluid (SAE nunber or military or Federal specification nutier)
Type of filter

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the shearing machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein.

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Except as otherwise,, specified in the contract or purchase order, the contractor may use "'his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the *Governnwit.* The Governmmt reserves the right to perform any inspections set forth in the specification where such inspections are deemd necessary to ensure supplies and services conform to prescribed require'nmts.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Governnmt for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed ~n accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, lt shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The shearing machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be a~plied to each shearinq machine prior to being offered for acceptance unde~" the contract. Unless-otherwise specified (see-6.2), quality conformance inspection shall consist of the examination in 4.6, the operational test in 4.7.2, the performance test in 4.7.3, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each shearing machine shall be examined to determine compliance with all requirenmts of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performd in an indoor fac lity with anbient conditions of 41 degrees to 104 degrees Fahrenheit (F) and *20* percent to 95 percent relative humidity.

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4.7.2 Operational test. Each shearing machine shall be operated at no load for not less han 30 minutes. Proper operation of all controls, rotors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Performance tests. The machine shall be tested for conformance with the requlrenmts of 3.6 by shearing mild steel of 60,000 psi tensile strength. The instrunmts and methods used in performing the tests shall be at the option of the contractor, provided they accurately determine the shearing capacity of the machine.

4.7.3.1 Shearing capacity. The shearing capacity of the machine shall be verified by shearing not less than six workplaces of maxirrum thickness and width as specified in tables I and II.

4.7.3.2 Shearing accuracy. The shearing accuracy of the machine shall be verified by shearing not less than six workplaces of maxirrum thickness and width as specified in tables I and II. The machine shall perform di,mnsional and parallel shearing to coq)ly with the requirenmts of 3.6.1.

4.7.3.3 Additional performance tests. Performance tests required in addition to or In lieu of the requirenwts of 4.7.3 shall be as specified and fully described (see 6.2).

4.7.4 Noise level test. Noise level measurements shall be made to ensure conformance with 3.2.5.2. Unless otherwise specified (see 6.2), the measurements shall be made at the operators work station and each corner of the shearing machine. Noise emitted by the shearing machine when operating under a no load condition shall not exceed 84 db when masured on the A scale of a type 1 or type 2 sound level inter at slow response. The test shall be conducted with the machine operating at intermediate and maxinum speeds.

4.7.5 Environmental protection certification of conformance. The contractor shall certify and malntaln substantlatlna evidence that the DrOdIJCt offered co~lies with all environmental protection ~estrictions as requ~red in 3.2.8. The governrmmt reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise stated under the provisions of the contract,

4.7.6 Electromagnetic interference control test. Equipment requiring electromagnetic Interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirewnts of section 5.

5. PACKAGING

5.1 Packaging requirenmts. The requirements for packaging shall be in accordance with MIL-ki-18058.

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6. NOTES , :

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The shearing machines covered by this specification are intended for use In metal shearing and squaring operations that are generally required in all phases of light gauge sheet metal fabrication work.

6.2 Acquisition requirements. Acquisition docuumts should specify the following:

a. Title, nutier, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual docunmts referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Systemof units for measuring and indicating devices (inch-pound) (see 3.2.1).

f. Additional safety and health requirermts, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5.1).

h. Noise level, if different (see 3.2.5.2).

i. Environmental protection requirenmts exceeding EPA/OSHA requirements, if different (see 3.2.8).

j. Lubrication system, if different (see 3.2.9).

k. Specify surface finish requirements, if different (see 3.3.3).

1. Painting, if different (see 3.3.5).

m. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

n. If machine drive of the Type I machine is to be a specific type, state required type (see 3.4.6).

o. Back gauge furnished with workpiece support mechanism, if required (see 3.4.7.1).

p* Side extension squaring gauge, if required (see 3.4.7.3).

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~. Light beam shearing gauge, if required (see 3.4.7.4).

r. Extra set of shear blades,if required (see 3.4.8).

s. Voltage requiremmts, if different (see 3.4.9).

t. Initial voltage, as specified (see 3.4.9).

u. Motors, if different (see 3.4.9.1).

v. Hour meter, if required (see 3.4.10).

w. Size and capacity, if different (see 3.5).

x. Specify physical size limitation, if required (see 3.5.1).

Y* Additional performance features, if required (see 3.6..2).

z. Specify and fully describe optional equipmnt, if required (see 3.8).

aa. Specify applicable repair parts, if required (see 3.9).

bb. Lubrication plate information, if different (see 3.10.1).

cc. Nameplate, if different (see 3.10.2).

dd. Hydraulic fluid requirenmts plate, if required (see 3.10.3).

ee. Hydraulic fluid plate information, if different (see 3.10.3).

ff. Inspection conditions, if different (see 4.3).

~9. First article inspection, if different (see 4.4).

hh. Quality conformance inspection, if different (4.5).

ii. Additional performance tests, if required (see 4.7.3.3).

jj. Noise level acceptance test, if different (see 4.7.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIOS) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirenmts of the specification acquisition. To ensure correct contractual application of the data requirermnts, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplemnt 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	1710 Nunber	010 Title	Suggested Tailoring
N/A	D1-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplenmtal Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Managenmt Systems and Data Requirements Control List (AMSDL), nust be researched to ensure that *only* current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sanple, a first production item, or a standard production item from the contractor's current inventory and the nunber of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition docurents regarding arrtingenmts for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Governnr?nt reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely *on* such production or test, must furnish evidence with the bid that prior Governnnt approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound conmnly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International Systemof Units based on "Le Systeme International d'Unites (S1), " of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurenmt system. In this specification, all measurements, dinmsions, sizes, and capacities are given in inch-pound units. These measurements may be converted to mtric units through the use of the conversion factors and rrdhods specified in FED-STD-376.

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6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

MIL-S-80220A	MIL-S-80220B
Type I	Type I
Size 1	Deleted
Size 2	Deleted
Size 3	Deleted
Size 4	Deleted
	Size 5 (New)
	Size 6 (New)
TyPe 11	Size 7 (New)
	Size 8 (New)
Size 1 (No change)	Size 9 (New)
	Size 10 (New)
	Size 11 (New)
	Size 12 (New)
	Size 13 (New)
	Size 14 (New)
	Size 15 (New)
	Size 16 (New)

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

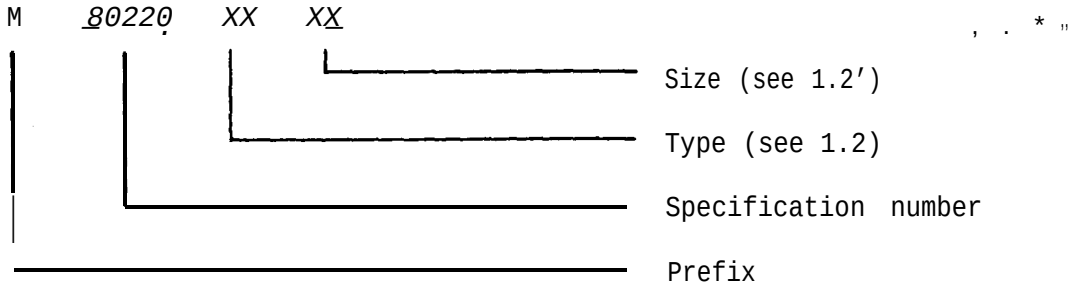
6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new maciline into its future environment.

6.10 Trainin Training required by the procuring activity should be provided as specl led in the contract.

6.11 Warrant Warranty requirements should be as specified by the procuring activity in the contract.

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6.12 Part or Identifying Nu~er (plN). The pIN to be used for shearin9 machines acquired to lis specifications are created as follows:



6.14 Subject term Key word listing.

Cutting
Metal working
Shearing
Slitting
Trimming

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

Preparing Activity:
DLA - 1P

Project (3445-0117)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5,6, and 7.
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I~RECOM"MENti:A.": cfi+,N"G~: .':	'" °°;:::7-;;?;(%	2. OOCUhlENT OATE (YYMMDO) 91-04-09

1. OOCUMENT TITLE
SHEARING MACHINES , SHEET METAL, POWER DRIVEN

NATURE OF CHANGE (/dent/fy paragraph number ● nd include proposed rewrite, if possible. Artach eara sheers ● s needed)

,. REASON FOR RECOMMENDATION

5. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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a. NAME	b. TELEPHONE (/nc/ude Area Code)	
Defense Industrial Plant Equipment Center ATTN : DIPEC-SSG	(1) Commercial (9ol)-775-4982	(2) AUTOVON 683&982
c. ADORE SS (Jrrc/ude Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defeme Quality and Standardization Off Ice 5203 Leesburg Pike. Su!te 1403. Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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